

Alex Deng

Senior Data Engineer

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SUMMARY

Senior Data Engineer with 12 years of experience designing and delivering scalable data solutions within the Microsoft ecosystem. Expertise in Azure Data Factory, Databricks, Synapse Analytics, and advanced SQL for building robust data pipelines and analytics platforms. Proven ability to lead architecture, implement data lakehouse strategies, and mentor teams while aligning solutions to business objectives. Strong background in data modeling, API ingestion, and enterprise data integration for high-impact analytics.

SKILLS

- **Data Engineering:** Azure Data Factory, Azure Databricks, Azure Synapse Analytics, Azure SQL Database, ADLS, Microsoft Fabric, Apache Spark
 - **Programming:** Python, SQL, T-SQL, Scala, PowerShell
 - **Data Modeling:** Dimensional Modeling, Star Schema, Data Warehousing, Data Lakehouse, ETL, ELT
 - **Cloud & Integration:** Azure, REST APIs, OAuth 2.0, JSON Processing, Data Integration, Data Migration
 - **Tools & Practices:** Git, CI/CD, Agile, Data Governance, Data Security, Metadata-driven Pipelines, Monitoring, Automation
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EXPERIENCE

Senior Data Engineer | BFC Software

04/2024 – Present

- Led architecture and implementation of enterprise-scale data pipelines using Azure Data Factory and Databricks, ensuring reliable ingestion, transformation, and delivery of high-volume business data.
- Designed and optimized dimensional data models using star schema techniques to support scalable reporting and analytics across multiple business domains and stakeholders.
- Developed metadata-driven data orchestration frameworks that improved pipeline reusability, maintainability, and reduced manual configuration efforts significantly.

- Integrated external data sources through REST APIs using OAuth 2.0 authentication while transforming complex JSON payloads into structured relational formats.
- Collaborated with cross-functional teams including analysts and cloud engineers to align technical solutions with business goals and reporting requirements.
- Implemented data lakehouse architecture using ADLS and Synapse Analytics to unify structured and semi-structured data for enterprise analytics use cases.
- Optimized complex SQL queries and Spark transformations to improve performance and reduce execution time across large-scale distributed data workloads.
- Established data quality and validation frameworks to ensure accuracy, completeness, and consistency across ingestion and transformation processes.
- Mentored junior engineers by providing technical guidance on Azure tools, best practices, and scalable data architecture design patterns.
- Led technical design sessions and documentation efforts to standardize development practices and improve knowledge sharing across teams.
- Implemented monitoring and alerting mechanisms to proactively identify failures and ensure high availability of critical data pipelines.
- Managed end-to-end delivery of data engineering projects including planning, execution, and stakeholder communication in a fast-paced environment.

Senior Data Engineer | Hibu

09/2022 – 03/2024

- Engineered scalable data pipelines using Azure Data Factory and Spark to process large volumes of marketing and customer analytics data efficiently.
- Designed and implemented star schema data models to support reporting systems and improve query performance for business intelligence teams.
- Developed robust ETL processes that transformed raw data into curated datasets for analytics and decision-making purposes.
- Integrated multiple third-party APIs and normalized JSON data into relational structures to support downstream reporting and analytics.
- Collaborated with business analysts to gather requirements and translate them into scalable data engineering solutions aligned with business needs.
- Optimized SQL queries and indexing strategies to enhance performance of analytical workloads on Azure SQL Database platforms.
- Implemented data governance practices including data validation, lineage tracking, and security compliance across data pipelines.
- Built reusable data transformation frameworks that reduced development time and improved consistency across multiple projects.
- Maintained and enhanced data lake solutions using ADLS to support structured and semi-structured data storage requirements.
- Participated in Agile development cycles and contributed to sprint planning, code reviews, and continuous improvement initiatives.

- Monitored pipeline performance and resolved production issues proactively to ensure minimal downtime and reliable data delivery.
- Documented system architecture and data flow processes to improve transparency and maintainability of data engineering solutions.

Data Engineer | Finra

09/2020 – 09/2022

- Developed data pipelines for regulatory data processing using Azure services, ensuring compliance with financial industry standards.
- Designed relational and dimensional data models to support reporting and analytics for compliance and risk management teams.
- Built ETL workflows to ingest, transform, and load large datasets from multiple internal and external sources.
- Implemented secure data integration practices including authentication protocols and encryption for sensitive financial data.
- Optimized SQL queries and database performance for high-volume transactional and analytical workloads.
- Collaborated with stakeholders to define data requirements and deliver scalable data solutions aligned with regulatory needs.
- Managed data quality checks and validation processes to ensure accuracy and integrity of critical financial datasets.
- Maintained data warehouse solutions supporting reporting and auditing requirements across business units.
- Implemented monitoring and logging solutions to track pipeline performance and troubleshoot issues effectively.
- Documented data architecture and processes to support audit requirements and ensure knowledge transfer across teams.

Data Engineer | Columbia University

02/2017 – 09/2020

- Designed and developed data pipelines to support academic research and administrative analytics using modern data engineering practices.
- Built ETL workflows to process structured and semi-structured data from various institutional systems and external sources.
- Developed dimensional models to enable reporting and analytics for academic and operational stakeholders.
- Collaborated with researchers and analysts to translate data requirements into scalable technical solutions.
- Optimized database queries and data transformations to improve performance and reduce processing time.

- Implemented data validation and quality checks to ensure reliability and consistency of research datasets.
- Maintained data storage solutions using relational databases and data lakes for diverse data types.
- Supported data integration efforts across multiple systems and departments within the university ecosystem.
- Documented data workflows and system architecture to support long-term maintenance and scalability.
- Participated in Agile development processes and contributed to continuous improvement initiatives.

Junior Data Engineer | dentsu

08/2012 - 01/2017

- Developed data processing workflows for marketing analytics using SQL and scripting tools to support campaign performance analysis.
- Built ETL pipelines to collect and transform advertising data from multiple sources into centralized reporting systems.
- Assisted in designing data models to support business intelligence and reporting needs across marketing teams.
- Optimized SQL queries to improve performance and reduce processing time for large-scale datasets.
- Collaborated with analysts to understand data requirements and deliver accurate and timely data solutions.
- Maintained data integrity through validation checks and error handling within ETL processes.
- Supported integration of third-party data sources and ensured consistency across datasets.
- Participated in system testing and debugging to ensure reliability of data processing workflows.
- Documented data processes and workflows to support knowledge sharing and maintainability.
- Contributed to continuous improvement efforts by identifying inefficiencies and proposing enhancements to data systems.

EDUCATION

Bachelor of Engineering (B.E.), Applied Mathematics and Economics - Stony Brook University (2012)